

CASCADIA DEPARTMENT OF TRANSPORTATION

Office of ITS Engineering & Procurement

REVISED ISSUE – SUPERSEDES RFP-CDOT-2026-1187

REQUEST FOR PROPOSAL

Statewide Intelligent Transportation System (ITS) Modernization, TMC Upgrade & Cybersecurity Resilience Program

A multi-year, multi-phase statewide ITS infrastructure, software, and cybersecurity modernization program

RFP Number	RFP-CDOT-2026-1187-R1
Issue Date	May 4, 2026
Proposal Submission Deadline	June 19, 2026, 4:00 PM PT
Issuing Organization	Cascadia Department of Transportation (CDOT)
Category	Intelligent Transportation Systems — Statewide ITS & TMC Modernization
Estimated Contract Value	See Section 12 — Commercial Terms

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1. Introduction & Background

The Cascadia Department of Transportation (“CDOT” or “the Department”) operates and maintains a statewide Intelligent Transportation System (ITS) network spanning approximately 1,464 centerline miles of interstate and state highways, following the February 2027 opening of the SR-509 Port Neelan Extension. The network now includes 660 closed-circuit television (CCTV) cameras, 355 dynamic message signs (DMS), and 196 ramp metering locations, all connected to a Traffic Management Center (TMC) located in Cascadia City.

CDOT's current Advanced Traffic Management System (ATMS) software platform was deployed in 2011. The platform vendor has announced that extended support will end in 2027, and the platform no longer supports modern integrations, AI-based incident detection, or current cybersecurity standards. Following a December 2025 statewide cybersecurity assessment that identified critical vulnerabilities in ITS field device communications, CDOT is reissuing this Request for Proposal (RFP) to solicit qualified vendors to modernize its ATMS platform, refresh field device hardware across the network, upgrade the communications backbone connecting field devices to the TMC, and implement enhanced cybersecurity resilience measures.

This program will be delivered in three phases: a Pilot Corridor phase, a Regional Rollout phase, and a Statewide Rollout phase, as described in Section 9 (Project Timeline).

2. Definitions

Term	Definition
ITS	Intelligent Transportation System — the combined field devices, communications network, and software used to monitor and manage highway traffic.
ATMS	Advanced Traffic Management System — the central software platform used by TMC operators to monitor and control field devices.
TMC	Traffic Management Center — CDOT's central facility for statewide traffic monitoring and incident response.
DMS	Dynamic Message Sign — an electronic roadside sign used to display traveler information and alerts.
CCTV	Closed-Circuit Television — roadside cameras used for traffic monitoring and incident verification.
Ramp Meter	A traffic signal on a freeway on-ramp used to regulate the rate of vehicles entering the freeway.
511 System	CDOT's statewide traveler information phone and web service.
NIST	National Institute of Standards and Technology — issuer of the cybersecurity framework referenced in this RFP.
SLA	Service Level Agreement — a contractually defined performance commitment.
MP	Mile Post — the reference marker system used to identify highway locations.
SOC	Security Operations Center — the 24/7 monitoring function required under this revision. [NEW]

3. Project Objectives

- Replace the legacy ATMS platform with a modern, standards-based ITS software platform.
- Refresh field device hardware (CCTV, DMS, ramp meters) that has reached or is approaching end of life.

- Upgrade the communications backbone connecting field devices to the TMC to support higher-bandwidth video and data.
- Implement AI-based automated incident detection to reduce average incident response time by at least 20%.
- Integrate the ATMS platform with CDOT's statewide 511 traveler information system.
- Harden ITS network cybersecurity in alignment with the NIST Cybersecurity Framework.
- Establish 24/7 Security Operations Center (SOC) monitoring of the ITS communications network. [NEW]

4. Scope of Work

1. Design, procure, and deploy a replacement Advanced Traffic Management System (ATMS) software platform.
2. Replace 660 end-of-life CCTV cameras with new pan-tilt-zoom (PTZ) IP cameras across all corridors listed in Appendix E. [MODIFIED — was 620]
3. Replace or upgrade 355 dynamic message sign (DMS) controllers, retaining existing sign structures where feasible. [MODIFIED — was 340]
4. Modernize 196 ramp metering locations with adaptive, demand-responsive metering algorithms. [MODIFIED — was 180]
5. Upgrade the fiber and wireless communications backbone connecting all field devices to the TMC, including the new SR-509 corridor.
6. Integrate the new ATMS platform with CDOT's statewide 511 traveler information system.
7. Implement AI-based automated incident detection algorithms across all CCTV-covered corridors.
8. Provide TMC operator training for approximately 52 operators across three shifts. [MODIFIED — was 45]
9. Harden ITS network cybersecurity in alignment with the NIST Cybersecurity Framework.
10. Establish 24/7 Security Operations Center (SOC) monitoring of the ITS communications network, including automated threat detection and incident response. [NEW]
11. Provide 5 years of post-deployment operations and maintenance (O&M) support.

5. Technical Requirements

5.1 ATMS Software Platform

- Support for NTCIP 1200-series standards for field device communication.
- Map-based operator console showing real-time status of all field devices.
- Configurable incident management workflow with automated response plan suggestions.
- Role-based access control aligned to TMC operator, supervisor, and administrator roles.

5.2 CCTV Cameras

- Minimum 4K (2160p) resolution with 30x optical zoom. [MODIFIED — was 1080p]
- Integrated video analytics for stopped-vehicle and wrong-way-driver detection.
- IP66-rated enclosures rated for the Department's operating climate range.

5.3 Dynamic Message Signs (DMS)

- Full-matrix LED display, minimum 3-line message capability.
- Solar-compatible power option for remote sign locations.
- NTCIP 1203 compliant sign controllers.

5.4 Ramp Metering

- Adaptive metering algorithms responsive to real-time mainline volume and occupancy.
- Local override capability for TMC operators during incidents.

5.5 Communications Network

- Fiber-optic backbone with a minimum 25 Gbps core ring capacity. [MODIFIED — was 10 Gbps]
- Cellular or licensed-wireless backup connectivity for remote field device locations.
- Network redundancy such that no single fiber cut disables more than 5% of connected field devices.

5.6 Hosting, Availability & Security

- Deployment model: vendor-hosted cloud environment only. On-premise TMC hosting will not be considered. [MODIFIED — REMOVES ON-PREM OPTION]
- System uptime SLA of 99.9% or higher for the ATMS software platform, measured monthly. [MODIFIED]
- Single sign-on (SSO) support via SAML 2.0, integrated with CDOT's existing Active Directory.
- Encryption at rest (AES-256 or equivalent) and in transit (TLS 1.2 or higher).
- Video and event data retention of a minimum of 4 years. [MODIFIED — was 2 years]

5.7 Cybersecurity

- Network architecture aligned with the NIST Cybersecurity Framework (Identify, Protect, Detect, Respond, Recover).
- Network segmentation isolating ITS field device traffic from CDOT's corporate IT network.
- Annual third-party penetration testing of the ATMS platform and communications network.
- 24/7 Security Operations Center (SOC) monitoring with a 30-minute notification SLA for confirmed intrusions. [NEW]
- Quarterly vulnerability scanning of all internet-facing ITS components. [NEW]

5.8 System Integrations

- Integration with CDOT's statewide 511 traveler information system for automated DMS message posting.
- API-based data feed to CDOT's public open-data portal for anonymized traffic volume data.
- Integration with the Department's existing computer-aided dispatch (CAD) system for incident data exchange.

6. Functional Requirements Matrix

Vendors must indicate compliance with each functional requirement in their proposal response using the codes: F (Fully Compliant), P (Partially Compliant — explain), C (Customization Required), N (Not Supported).

ID	Requirement	Priority
FR-01	Real-time map-based device status console	Mandatory
FR-02	Automated incident detection from CCTV video analytics	Mandatory
FR-03	Configurable incident response plan library	Mandatory
FR-04	Automated DMS message posting from approved response plans	Mandatory
FR-05	Adaptive ramp metering with manual override	Mandatory
FR-06	Historical event and video data archive with search	Mandatory
FR-07	Role-based dashboard customization	Mandatory
FR-08	Mobile app for field technician device diagnostics	Mandatory
FR-09	Automated device health monitoring with predictive maintenance alerts	Desired
FR-10	Public open-data API for traffic volume exports	Mandatory
FR-11	511 system integration for automated traveler alerts	Mandatory
FR-12	CAD system integration for incident data exchange	Mandatory
FR-13	Automated SOC alert correlation with ATMS incident workflow	Mandatory
FR-14	Anomalous network traffic detection and alerting	Mandatory

7. Eligibility Criteria

To be eligible for consideration, vendors must meet all of the following minimum criteria:

- Minimum of 10 years of experience providing ITS/ATMS systems integration services to state or provincial transportation agencies. [MODIFIED]
- Minimum annual revenue of \$60,000,000 (audited financials for the most recent fiscal year required). [MODIFIED]
- At least 5 reference implementations for state or provincial DOT ITS deployments covering 500 or more field devices, completed within the past 7 years, of which at least 2 must include a 24/7 SOC monitoring component. [MODIFIED]
- A Professional Engineer (PE) licensed in the state, on staff, responsible for ITS system design.
- Vendor's SOC personnel must hold current CISSP or GIAC certification. [NEW]
- Minimum insurance coverage: \$15,000,000 general liability; \$8,000,000 professional liability/E&O; \$5,000,000 cyber liability. [MODIFIED]
- Performance bond equal to 100% of the awarded contract value.

8. Deliverables

#	Deliverable	Due
1	Project Charter & Implementation Plan	45 days after NTP
2	ATMS Platform System Design Document	90 days after NTP
3	Pilot Corridor Deployment (I-90, MP 0–15)	Per Phase 1 milestone
4	Pilot Corridor Performance Report	60 days after Phase 1 go-live
5	Regional Rollout Deployment Plan	Prior to Phase 2
6	Regional Rollout Deployment (5 corridors)	Per Phase 2 milestone
7	Statewide Rollout Deployment Plan	Prior to Phase 3
8	Statewide Rollout Deployment (remaining corridors, incl. SR-509)	Per Phase 3 milestone
9	Cybersecurity Architecture & Penetration Test Report	Prior to each phase go-live
10	SOC Monitoring Runbook & Escalation Procedures [NEW]	Prior to Phase 1 go-live
11	TMC Operator Training Materials & Delivered Sessions (52 operators)	Prior to each phase go-live
12	511 & CAD Integration Testing Report	Prior to statewide go-live
13	Final Documentation & As-Built Package	Project close-out

9. Project Timeline

Milestone	Date
RFP Reissue Date	May 4, 2026
Vendor Q&A Deadline	May 18, 2026
Proposal Submission Deadline	June 19, 2026
Shortlist Notification	July 3, 2026

Milestone	Date
Vendor Presentations	July 13 – 17, 2026
Contract Award	July 31, 2026
Notice to Proceed (NTP)	August 24, 2026
SOC Monitoring Go-Live [NEW]	January 2027
Phase 1 — Pilot Corridor Go-Live (I-90)	June 2027
Phase 2 — Regional Rollout Go-Live (5 corridors + SR-509)	March 2028
Phase 3 — Statewide Rollout Go-Live (remaining corridors)	December 2028
Full Project Completion	April 2029

10. Evaluation Criteria

Proposals will be scored on a 100-point scale as follows:

Criterion	Weight
Technical Solution & Architecture	20 points
Vendor Experience & References	15 points
Implementation Approach & Timeline	10 points
Cost	25 points
Cybersecurity Approach & SOC Design	20 points
O&M / Support Plan	10 points

11. Insurance & Bonding Requirements

- Commercial General Liability: \$15,000,000 per occurrence. [MODIFIED]
- Professional Liability / Errors & Omissions: \$8,000,000 per claim. [MODIFIED]
- Cyber Liability: \$5,000,000 per occurrence, covering data breach and business interruption. [NEW]
- Performance Bond: 100% of total contract value, issued by a surety rated A- or better by A.M. Best.
- CDOT and its officers, agents, and employees must be named as additional insureds on all policies except professional liability.

12. Commercial Terms

Estimated Project Budget: \$118,000,000 – \$148,000,000

Contract Term: 7 years, with two (2) optional 12-month renewals. [MODIFIED]

Payment Schedule

- 10% upon contract signing / NTP
- 10% upon System Design Document acceptance
- 10% upon SOC monitoring go-live [NEW]
- 20% upon Phase 1 Pilot Corridor go-live
- 25% upon Phase 2 Regional Rollout go-live
- 15% upon Phase 3 Statewide Rollout go-live
- 10% upon final acceptance

Other Commercial Terms

- Warranty period: 3 years following each phase's go-live, covering defect remediation at no additional cost.
- Payment terms: Net 30 from invoice date. [MODIFIED]
- Liquidated damages: 0.35% of total contract value per week of delay against the approved schedule, capped at 10% of total contract value. [MODIFIED]

13. Submission Instructions

Proposals must be submitted electronically in PDF format to its-proposals@cdot.state.gov no later than the deadline listed above. Proposals must remain valid for 180 days from the submission deadline. Late submissions will not be considered.

14. Terms & Conditions

14.1 Governing Law

This RFP and any resulting contract shall be governed by the laws of the State of Cascadia.

14.2 Confidentiality

All proposal materials submitted become the property of CDOT and are subject to applicable public records law, subject to any properly marked trade secret exemptions.

14.3 Reservation of Rights

CDOT reserves the right to reject any or all proposals, waive minor irregularities, and negotiate final contract terms with the selected vendor.

14.4 Intellectual Property

All custom configurations, integrations, and documentation developed specifically for CDOT under this engagement shall become the property of CDOT upon final payment. Vendor retains ownership of pre-existing, generally licensed software.

14.5 Indemnification

Vendor shall indemnify, defend, and hold harmless CDOT and its officers, agents, and employees from any claims, damages, or losses arising from the vendor's negligent acts or omissions in the performance of this contract.

14.6 Dispute Resolution

The parties shall attempt to resolve any dispute through good-faith negotiation before pursuing formal legal remedies. Unresolved disputes shall be subject to the jurisdiction of the courts of the State of Cascadia.

14.7 Force Majeure

Neither party shall be liable for delays caused by events beyond its reasonable control, including natural disasters, acts of government, or other force majeure events, provided the affected party gives prompt notice and uses reasonable efforts to mitigate the delay.

14.8 Assignment

Vendor may not assign or subcontract a material portion of this contract without CDOT's prior written consent.

14.9 Cybersecurity Incident Notification [NEW]

Vendor must notify CDOT's Chief Information Security Officer within 30 minutes of confirming any intrusion or anomalous activity on the ITS communications network, and must deliver a written incident report within 3 business days of resolution.

15. Contact Information

Issuing Officer	Marcus Alvarez
Title	Senior ITS Procurement Manager
Email	malvarez@cdot.state.gov
Phone	(360) 555-0209

Appendix A: Pricing Form

Vendors must complete the following pricing form in full. Unit prices must be all-inclusive of labor, materials, travel, and overhead unless otherwise noted.

Line Item	Description	Price
A-1	ATMS Software Platform Licensing (Statewide)	\$_____
A-2	ATMS Implementation & Configuration Services	\$_____
A-3	CCTV Camera Hardware & Installation (660 units)	\$_____
A-4	DMS Controller Hardware & Installation (355 units)	\$_____
A-5	Ramp Metering Upgrade (196 locations)	\$_____
A-6	Communications Backbone Upgrade (25 Gbps core)	\$_____
A-7	511 & CAD Integration Services	\$_____
A-8	Cybersecurity Architecture & Penetration Testing	\$_____
A-9	24/7 SOC Monitoring Setup & Annual Subscription [NEW]	\$_____
A-10	TMC Operator Training (52 operators)	\$_____
A-11	5 Years Post-Deployment O&M Support	\$_____
A-12	Total Proposed Price	\$_____

Appendix B: Reference Form

Provide at least 5 references meeting the eligibility criteria in Section 7, including at least 2 with a 24/7 SOC monitoring component. Duplicate this table as needed.

Field	Response
Client Agency	
Contact Name & Title	
Phone / Email	
Device Count / Corridor Miles	
Included SOC Monitoring? (Y/N)	
Contract Value	
Completion Date	

Appendix C: Compliance Certification Checklist

Requirement	Certified (Y/N)
Vendor meets minimum 10 years relevant experience	
Vendor meets minimum annual revenue threshold	
Vendor holds required insurance coverage	
Vendor can provide performance bond as specified	

Requirement	Certified (Y/N)
Vendor's ITS design lead is a licensed PE	
Proposed ATMS supports NTCIP 1200-series standards	
Proposed CCTV cameras meet minimum 4K resolution/zoom specification	
Proposed solution aligns with the NIST Cybersecurity Framework	
Vendor's SOC personnel hold CISSP or GIAC certification	
Proposed hosting is cloud-only with no on-premise TMC option	

Appendix D: Technical Specifications Detail

D.1 CCTV Camera Specification

Parameter	Minimum Requirement
Resolution	4K (2160p) [MODIFIED — was 1080p]
Optical Zoom	30x
Enclosure Rating	IP66
Low-Light Performance	0.01 lux color, 0.001 lux B/W
Video Analytics	Stopped-vehicle and wrong-way detection
Communication Protocol	ONVIF Profile S/T compliant

D.2 Dynamic Message Sign (DMS) Specification

Parameter	Minimum Requirement
Display Type	Full-matrix LED
Message Lines	3 lines minimum
Brightness	Auto-dimming, 1,000–10,000 nits range
Controller Protocol	NTCIP 1203 compliant
Power Option	AC and solar-compatible

D.3 Ramp Meter Controller Specification

Parameter	Minimum Requirement
Metering Algorithm	Adaptive, volume/occupancy responsive
Detector Interface	Loop and radar detector compatible
Override Capability	Local and remote TMC override
Communication Protocol	NTCIP 1207 compliant

D.4 Communications Network Specification

Parameter	Minimum Requirement
Core Backbone Capacity	25 Gbps minimum, ring topology [MODIFIED — was 10 Gbps]
Field Device Connectivity	Fiber preferred; licensed wireless where fiber infeasible
Redundancy	No single fiber cut disables more than 5% of devices
Backup Connectivity	Cellular failover for remote sites

D.5 Security Operations Center (SOC) Specification [NEW]

Parameter	Minimum Requirement
Monitoring Coverage	24/7/365
Intrusion Notification SLA	30 minutes from confirmed detection
Vulnerability Scanning Frequency	Quarterly, all internet-facing components
Personnel Certification	CISSP or GIAC required for SOC analysts

Appendix E: Corridor & Device Inventory List

The following table lists the major highway corridors included in the scope of this program, with existing device counts to be replaced or upgraded. Minor connector routes and short spur segments are not itemized individually here but are included in the statewide totals in Section 1; a complete site-by-site inventory will be provided to the selected vendor during the design phase. Vendors should use this list to develop phased deployment plans and cost estimates.

Corridor ID	Route	Segment Description	CCTV Cameras	DMS Units	Ramp Meters
C-01	I-90	MP 0–15, Cascadia City – Eastgate	18	9	12
C-02	I-90	MP 15–42, Eastgate – Summit Pass	14	7	0
C-03	I-5	MP 0–22, Port Neelan – Cascadia City	26	14	20
C-04	I-5	MP 22–58, Cascadia City – Fairhaven	38	19	24
C-05	I-5	MP 58–90, Fairhaven – Northgate Border	19	11	8
C-06	I-405	MP 0–17, Eastgate Loop	22	12	18
C-07	US-2	MP 0–35, Cascadia City – Timberline	12	6	0
C-08	US-2	MP 35–71, Timberline – Highpass	9	5	0
C-09	SR-522	MP 0–19, Northgate – Riverside	15	8	10
C-10	SR-167	MP 0–24, Valley Junction – Southport	17	9	18
C-11	I-82	MP 0–38, Southport – Vinewood	13	7	0
C-12	US-97	MP 0–45, Highpass – Dryland	8	4	0
C-13	SR-14	MP 0–28, Riverside – Gorgeview	11	6	0
C-14	I-205	MP 0–12, Southport Loop	16	9	12
C-15	SR-99	MP 0–16, Cascadia City Waterfront	20	11	0
C-16	SR-509	MP 0–14, Port Neelan Extension (opened Feb 2027) [NEW]	10	6	8

Total: 16 corridors — 268 CCTV cameras, 143 DMS units, 130 ramp meters.

Note: Corridor C-16 (SR-509 Port Neelan Extension) is new in this revision. Corridor C-04 (I-5, Cascadia City – Fairhaven) device counts were revised upward following a supplemental field survey completed in April 2026. Corridor C-10 (SR-167) ramp meter count was revised upward to reflect four newly constructed on-ramps.

Appendix F: Sample Service Level Metrics

The following illustrative service levels will inform the O&M support contract to be negotiated with the selected vendor following statewide go-live:

Metric	Target
ATMS platform uptime	99.9% monthly [MODIFIED — was 99.5%]
Critical field device fault response (on-site)	Within 6 business hours [MODIFIED — was 8 hours]
Non-critical field device fault response (on-site)	Within 3 business days
Software defect (Severity 1) resolution	Within 24 hours
Software defect (Severity 2) resolution	Within 5 business days
SOC intrusion notification [NEW]	Within 30 minutes of confirmed detection